**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Paper 1: Software, Impacts and Legislation

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How does encryption protect transmitted data?

2. What does FROM name in SQL?

A) The output columns
B) The filter condition
C) The source table
D) The sort order

3. "A syntax error is any incorrect answer produced by syntactically valid code." Correct this. Explain why.

4. A program repeats the same price calculation in several places. "Copying long code blocks always makes maintenance easier." Correct this. How would a subprogram help?

## Today's objectives

Explain the core ideas of paper 1: software, impacts and legislation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Operating-system services**

Manage memory, files, processes and peripherals.

**Utility tools**

Encryption, compression and HDD defragmentation serve different needs.

**Societal impacts**

Evaluate benefit and risk for identified stakeholders.

**Legal framework**

Match misuse, copyright and personal-data concerns to the appropriate legislation/rules.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Operating-system services** | Manage memory, files, processes and peripherals. |
| **Utility tools** | Encryption, compression and HDD defragmentation serve different needs. |
| **Societal impacts** | Evaluate benefit and risk for identified stakeholders. |
| **Legal framework** | Match misuse, copyright and personal-data concerns to the appropriate legislation/rules. |

## I Do - One worked example

**Evaluate a school cloud backup service in terms of benefits and risks.**

1. Off-site backups can support data recovery.
2. Internet dependency and availability can limit access.
3. Student records need protected access and appropriate lawful handling.
4. Justify using secure, tested backup procedures.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A staff member uploads copyrighted media without permission. Identify the issue.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Complete the Boolean expression that rejects collecting data without a purpose.**

```
hasPurpose = false
if ... then
print("Collect data")
else
print("Do not collect")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. Give one benefit of OS multitasking. [2 marks]

Q3. What is the main purpose of a compression utility? [2 marks]

Q4. Identify relevant law for accessing another user's account without permission. [1 mark]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Operating systems can manage more than one running process. ☐ T ☐ F
2. Manage memory, files, processes and peripherals. ☐ T ☐ F
3. Encryption, compression and HDD defragmentation serve different needs. ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key. ☐

**Q2** C - The source table ☐

**Q3** A syntax error breaks the language's rules. The affected code cannot be translated or run correctly. ☐

**Q4** Put shared code in one subprogram. It can be reused, tested and changed in one place. ☐

## Code First - Model

```
hasPurpose = false
if hasPurpose then
print("Collect data")
else
print("Do not collect")
endif
```

**Output: Do not collect**

## We Do - Check the model

1. Conclude possible copyright infringement. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Allows multiple applications/processes to make progress through scheduling. ☐

**Q3** Reduce the size of stored/transmitted data. ☐

**Q4** Computer Misuse Act 1990. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**